

Continued development of computer-vision based feature extraction workflows for multiple GIS assets

Charles Major
Manager of GIS Data Engineering
RIDOT 2026 Symposium on AI in the Road & Mobility Sector



We needed to do things more efficiently

- MassDOT business units needed higher quality and new forms of complex and nuanced information
- Update cadence became paramount
- Business strategy revealed synergistic value to these additional layers
- Traditional data development methods were simply too slow and costly to generate data at the volume and quality needed
- Previous GIS extraction success is leading us to expansion of asset collection types and avenues



MassDOT's GeoAI Journey

- Crosswalk Facilities
- Pedestrian Inventory
- CCTVs
- MassDOT owned parcel footprints
- Digitization of Assets from PDFs and paper maps
- Sign Extraction from imagery and video logs
- Rail right of Way extraction from engineering design plans
- Pavement marking extractions for better modeled turn exclusions (Upcoming)



Crosswalks

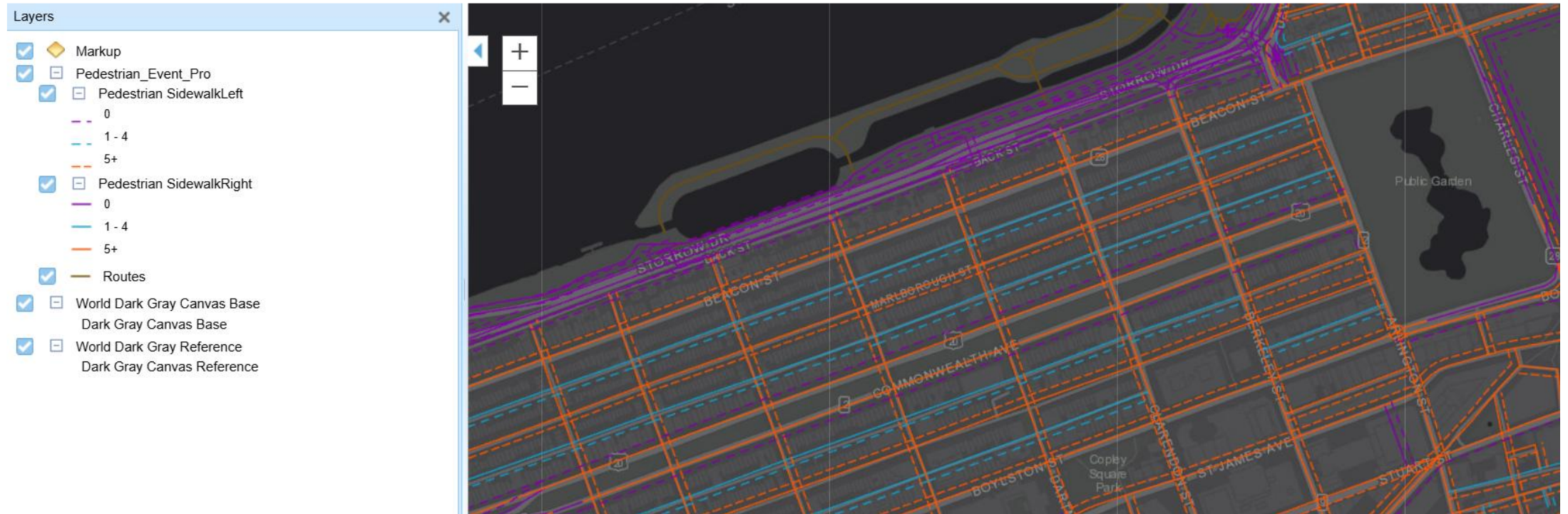
- MassDOT and Research Groups at UMass partnered to get a statewide inventory of all crosswalks for the first time.
- Modified Yolo V8 configured to use bounding boxes instead of free form shapes
- Further work to ordinally classify based on crosswalk degradation for safety analysis and project prioritization.



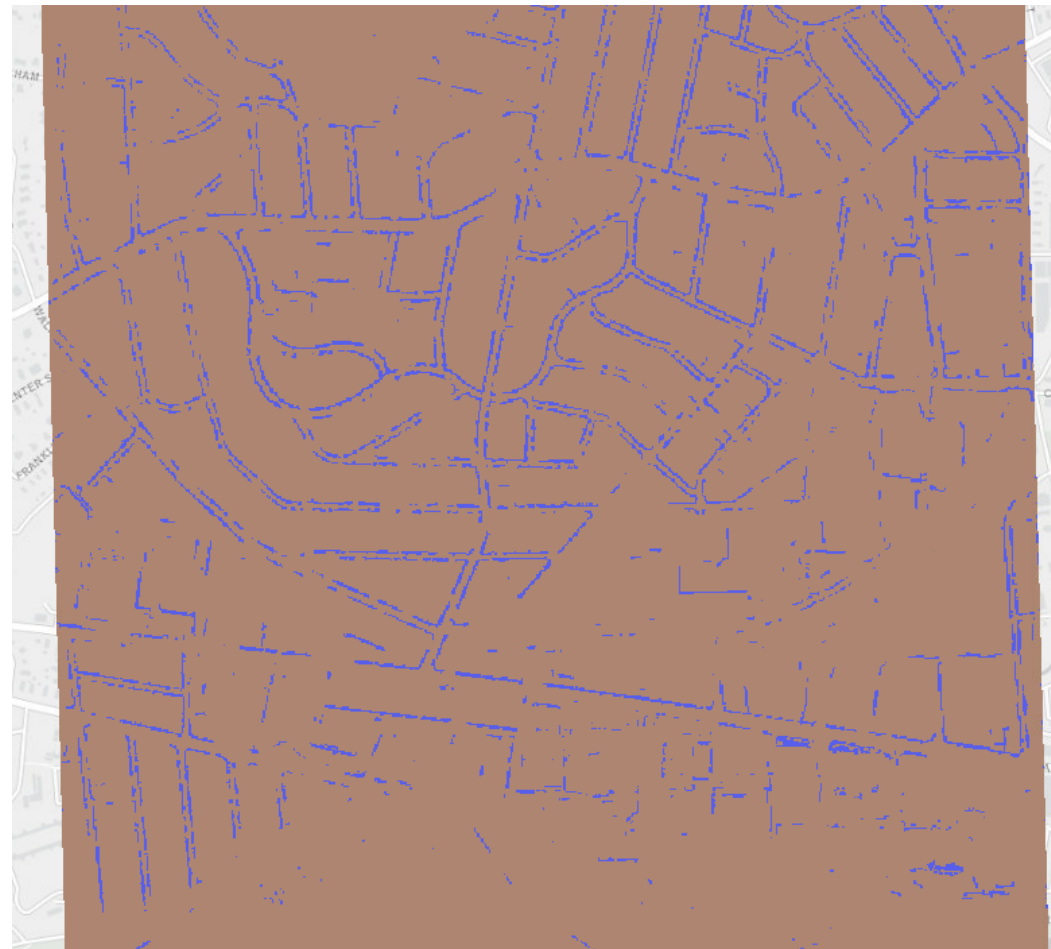
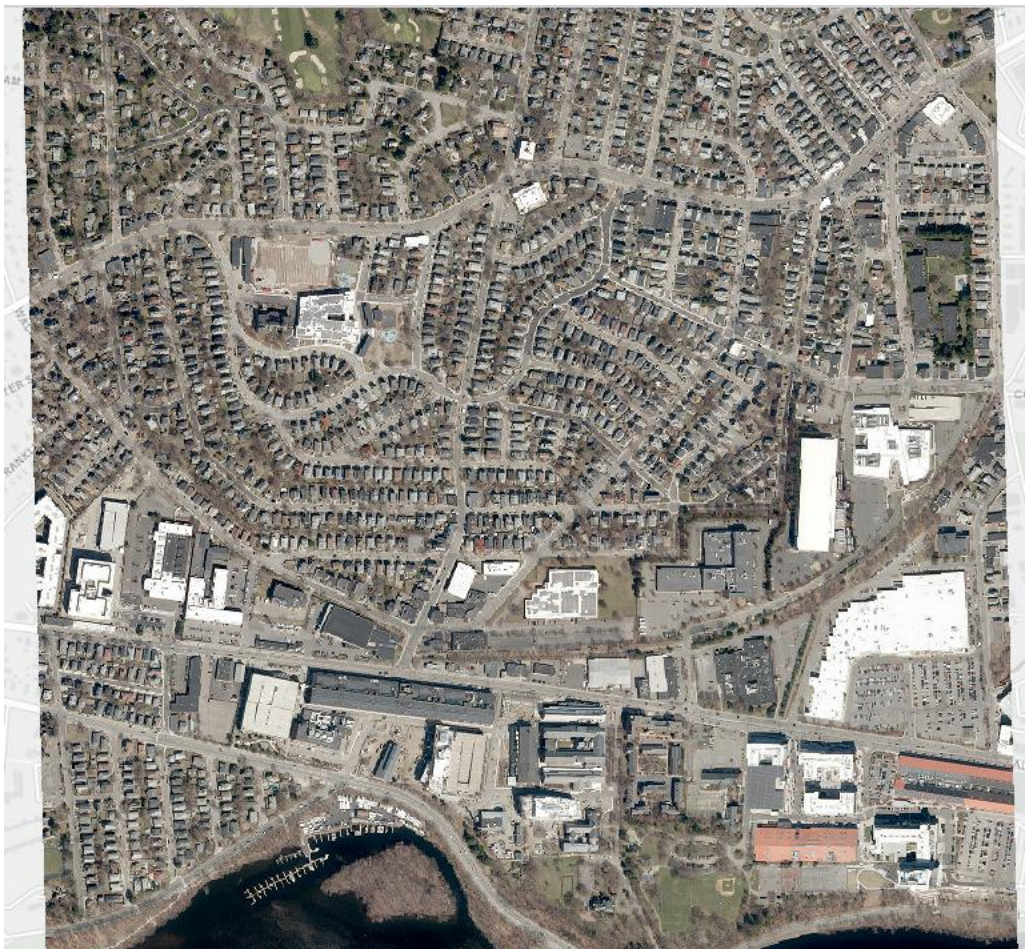
Dr. Xie Umass Lowell



Pedestrian Inventory and sidewalk classification



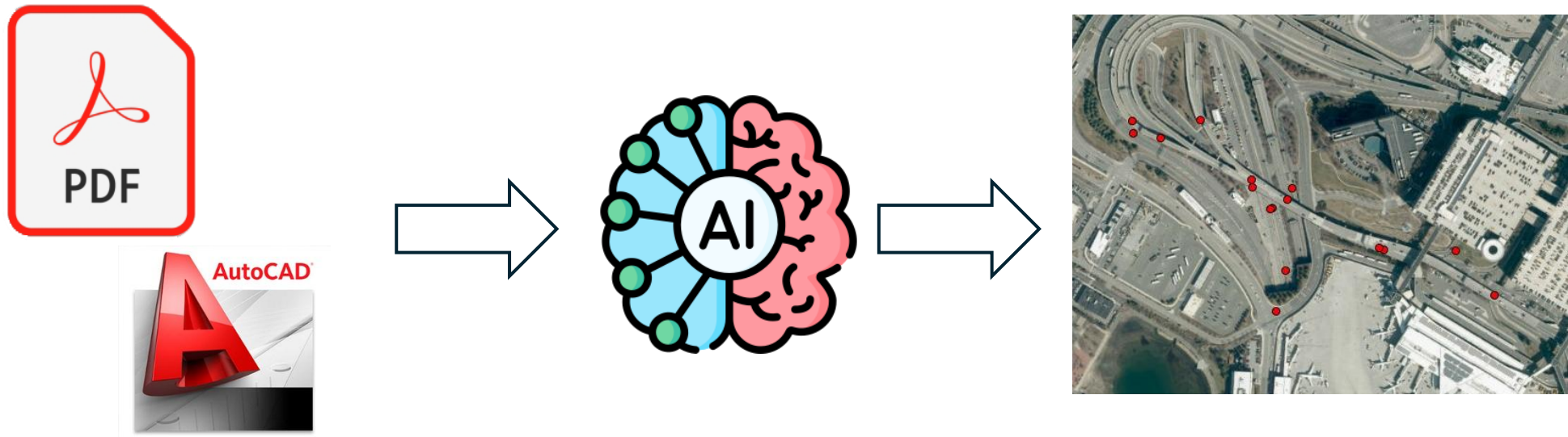
Sidewalk Classification

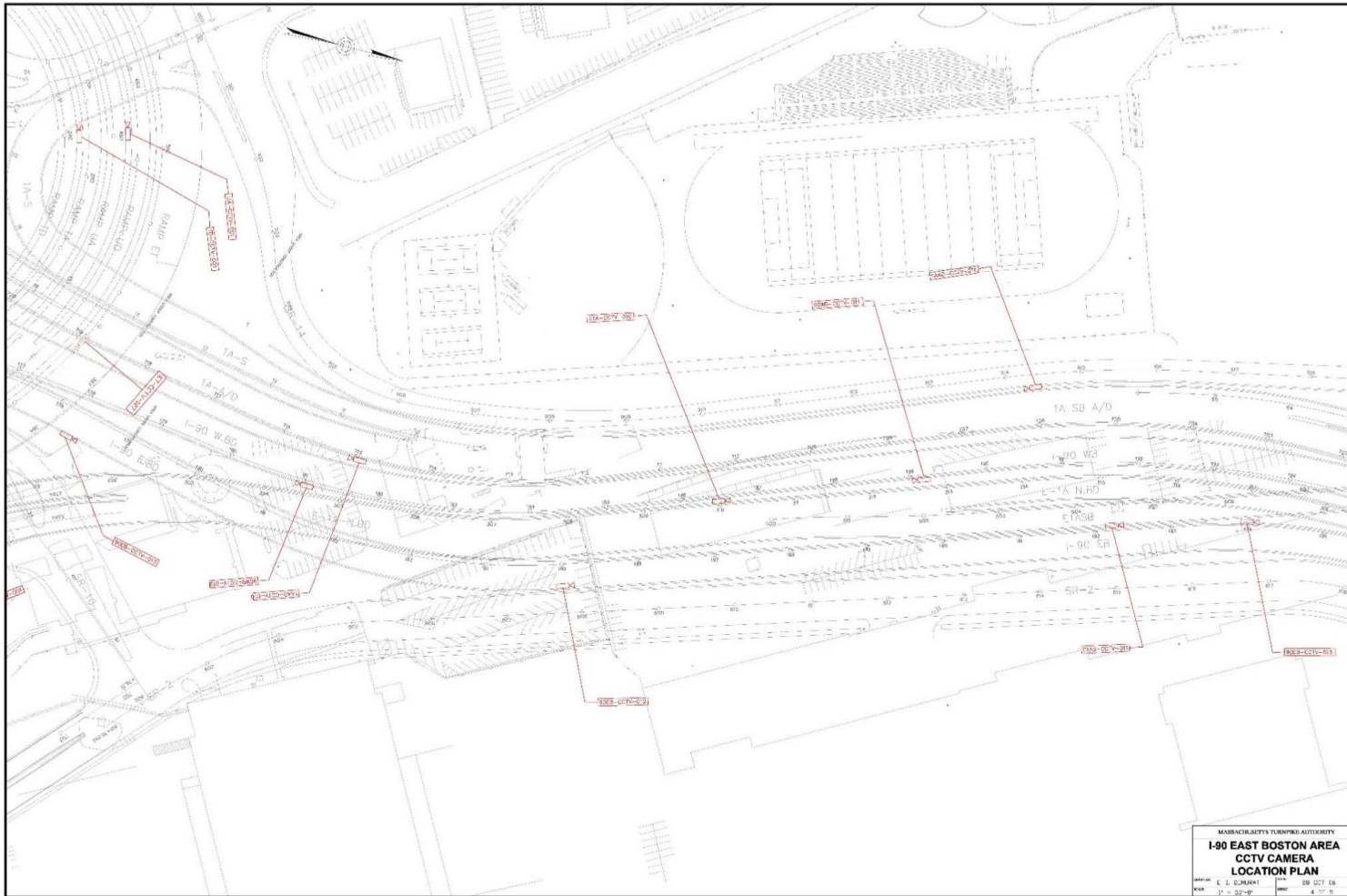


MassDOT Facilities footprint and Solar Feasibility



Convert documents to GIS Data

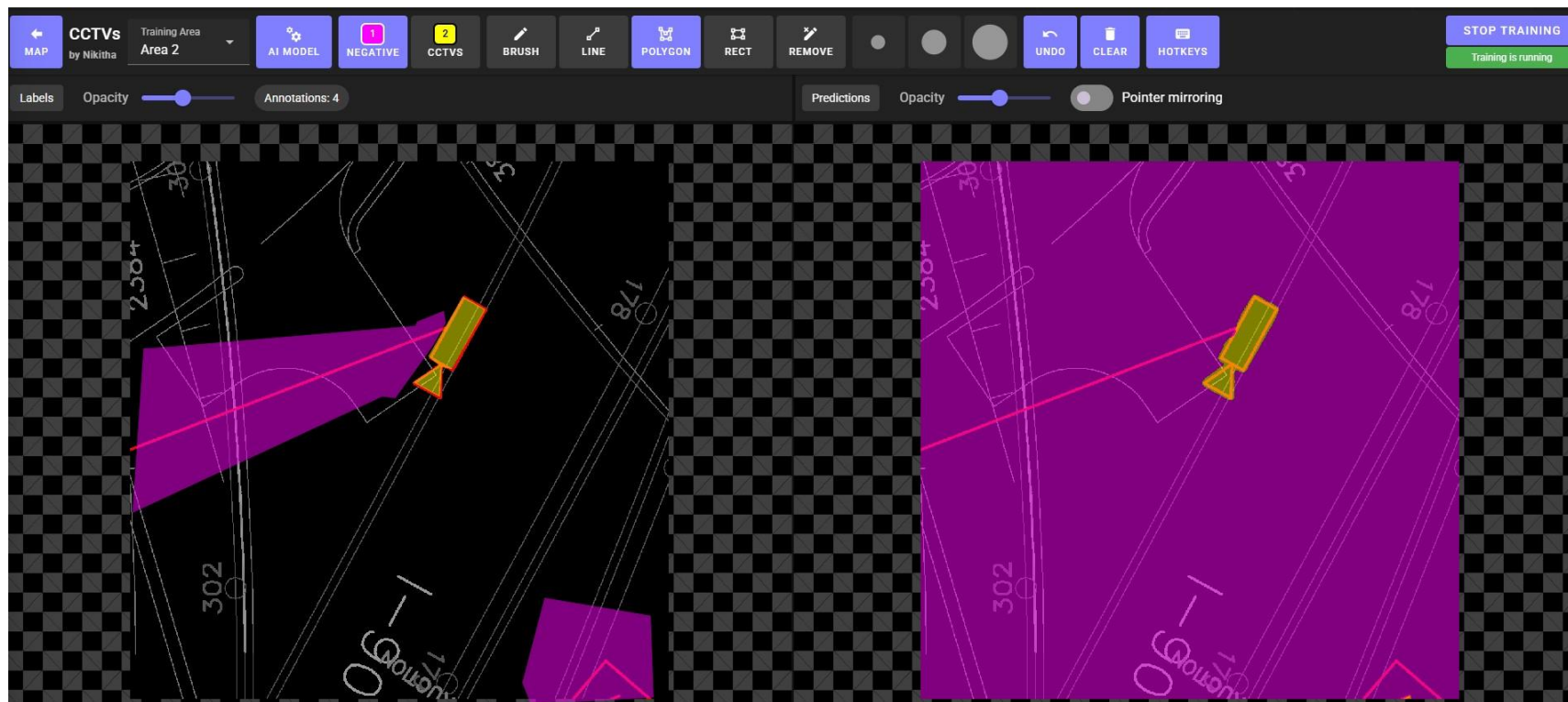




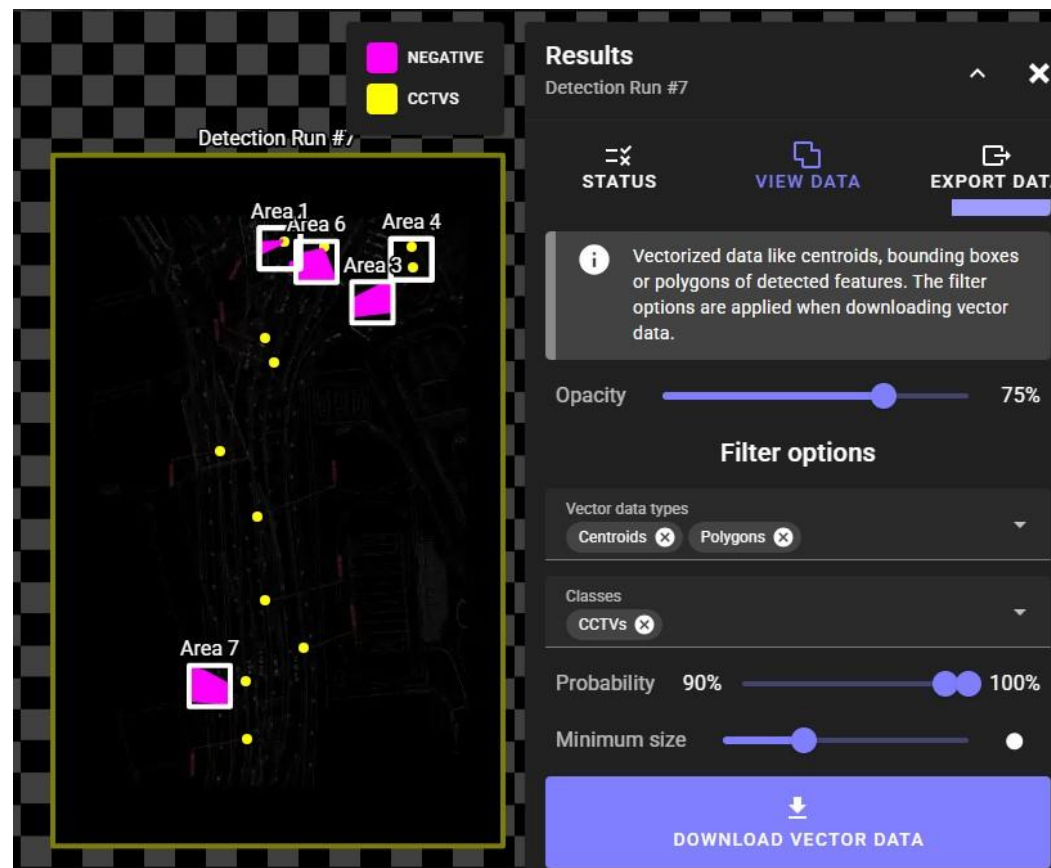
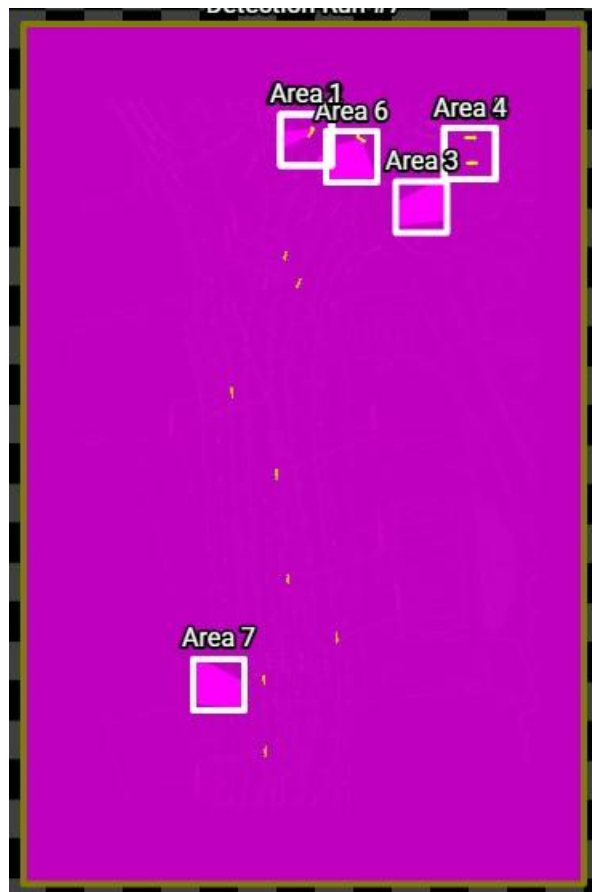
	A	B	C	M	N	O	P	Q	R	S	T
1	X	B	C	.	K	(ALT Sort)	.	.	(PROPOSED, add in MM)	.	.
2	Asset Equipment ID #	HOC Equipment ID #	HOC Monitor Display Nomenclature (50 char max.)	District	Description (free text)	TrafficLand Web and VMC Description	Chg Coun	OK??	TrafficLand Web & VMC, w/ MM# info	Chg Coun	OK??
118	6WB-CCTV--68.2	6MM68.2WB--	006-MM068.2WB--Barnstable-Ex 68 @ Lyannough Rd	5	Ex 68 @ Lyannough Rd	Rt6-WB-Barnstable-Ex 68 @ Lyannough Rd	39	<50	Rt6-WB-MM68.2-Barnstable-Ex 68 @ Lyannough Rd	46	<50
122	6EB-CCTV--82.0	6MM82.0EB--	006-MM082.0EB--Harwich-After Ex 82 @ Pleasant Lake Ave	5	After Ex 82 @ Pleasant Lake Ave	Rt6-EB-Harwich-After Ex 82 @ Pleasant Lake Ave	47	<50	Rt6-EB-MM82.0-Harwich-After Ex 82 @ Pleasant Lake Ave	54	Reduce text!
479	95SB-CCTV--0.4	95MM0.4SB--	095-MM000.4SB--Attleboro-Before Ex 1 @ Washington St.	5	Before Ex 1 @ Washington St.	I-95-SB-Attleboro-Before Ex 1 @ Washington St.	46	<50	I-95-SB-MM0.4-Attleboro-Before Ex 1 @ Washington St.	52	Reduce text!
480	95NB-CCTV--1.1	95MM1.1NB--	095-MM001.1NB--Attleboro-Before Ex 2B @ Newport Ave	5	Before Ex 2B @ Newport Ave	I-95-NB-Attleboro-Before Ex 2B @ Newport Ave	44	<50	I-95-NB-MM1.1-Attleboro-Before Ex 2B @ Newport Ave	50	<50
483	95SB-CCTV--4.2	95MM4.2SB--	095-MM004.2SB--Attleboro-Before 'Ex 4A @ South Ave	5	Before 'Ex 4A @ South Ave	I-95-SB-Attleboro-Before 'Ex 4A @ South Ave	43	<50	I-95-SB-MM4.2-Attleboro-Before 'Ex 4A @ South Ave	49	<50
484	95NB-CCTV--4.2	95MM4.2NB--	095-MM004.2NB--Attleboro-After Ex 4 @ South Ave	5	After Ex 4 @ South Ave	I-95-NB-Attleboro-After Ex 4 @ South Ave	40	<50	I-95-NB-MM4.2-Attleboro-After Ex 4 @ South Ave	46	<50
553	195EB-CCTV--1.4	195MM1.4EB--	195-MM001.4EB--CCTV-Seekonk-East of Ex 1 @ Anthony St	5	East of Ex 1 @ Anthony St	I-195-EB-Seekonk-East of Ex 1 @ Anthony St	42	<50	I-195-EB-MM1.4-Seekonk-East of Ex 1 @ Anthony St	48	<50
556	195WB-CCTV--6.8	195MM6.8WB--	195-MM006.8WB--CCTV-Swansea-West of Ex 8 @ Maple Ave	5	West of Ex 8 @ Maple Ave	I-195-WB-Swansea-West of Ex 8 @ Maple Ave	41	<50	I-195-WB-MM6.8-Swansea-West of Ex 8 @ Maple Ave	47	<50
560	195WB-CCTV--9.7	195MM9.7WB--	195-MM009.7WB--CCTV-Somerset-After Ex 10A @ Lees River Ave	5	After Ex 10A @ Lees River Ave	I-195-WB-Somerset-After Ex 10A @ Lees River Ave	47	<50	I-195-WB-MM9.7-Somerset-After Ex 10A @ Lees River Ave	53	Reduce text!
561	195WB-CCTV--9.9	195MM9.9WB--	195-MM009.9WB--CCTV-Somerset-Ex 10A @ Wilbur Ave	5	Ex 10A @ Wilbur Ave	I-195-WB-Somerset-Ex 10A @ Wilbur Ave	37	<50	I-195-WB-MM9.9-Somerset-Ex 10A @ Wilbur Ave	43	<50
563	195EB-CCTV--12.7	195MM12.7EB--	195-MM012.7EB--CCTV-Fall River-Ex 13 @ Plymouth Ave	5	Ex 13 @ Plymouth Ave	I-195-EB-Fall River-Ex 13 @ Plymouth Ave	40	<50	I-195-EB-MM12.7-Fall River-Ex 13 @ Plymouth Ave	47	<50
567	195EB-CCTV--16.4	195MM16.4EB--	195-MM016.4EB--CCTV-Westport-Ex 16 @ Route 88	5	Ex 16 @ Route 88	I-195-EB-Westport-Ex 16 @ Route 88	34	<50	I-195-EB-MM16.4-Westport-Ex 16 @ Route 88	41	<50
568	195WB-CCTV--16.4	195MM16.4WB--	195-MM016.4WB--CCTV-Westport-Ex16 @ Route 88	5	Ex16 @ Route 88	I-195-WB-Westport-Ex16 @ Route 88	33	<50	I-195-WB-MM16.4-Westport-Ex16 @ Route 88	40	<50

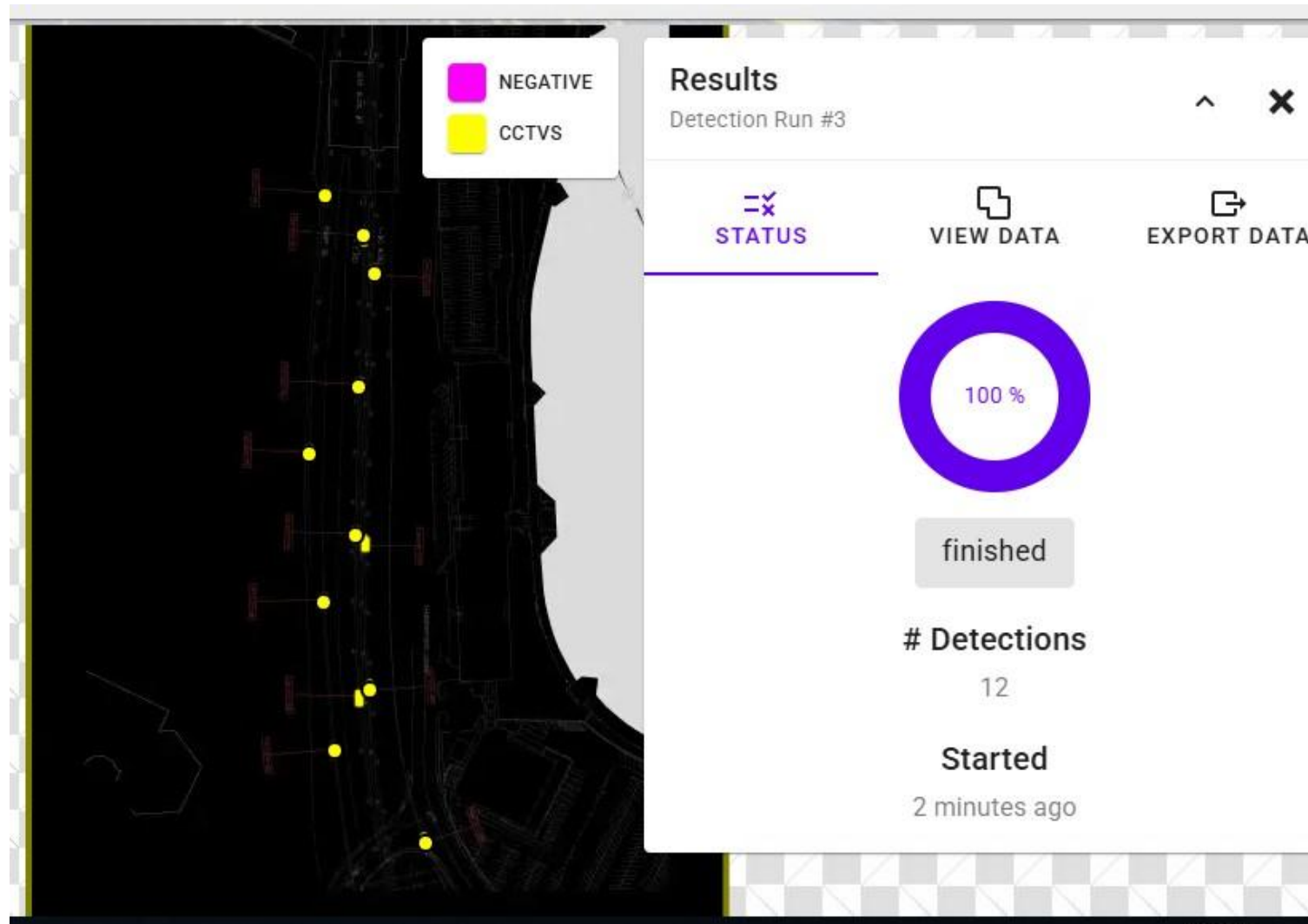


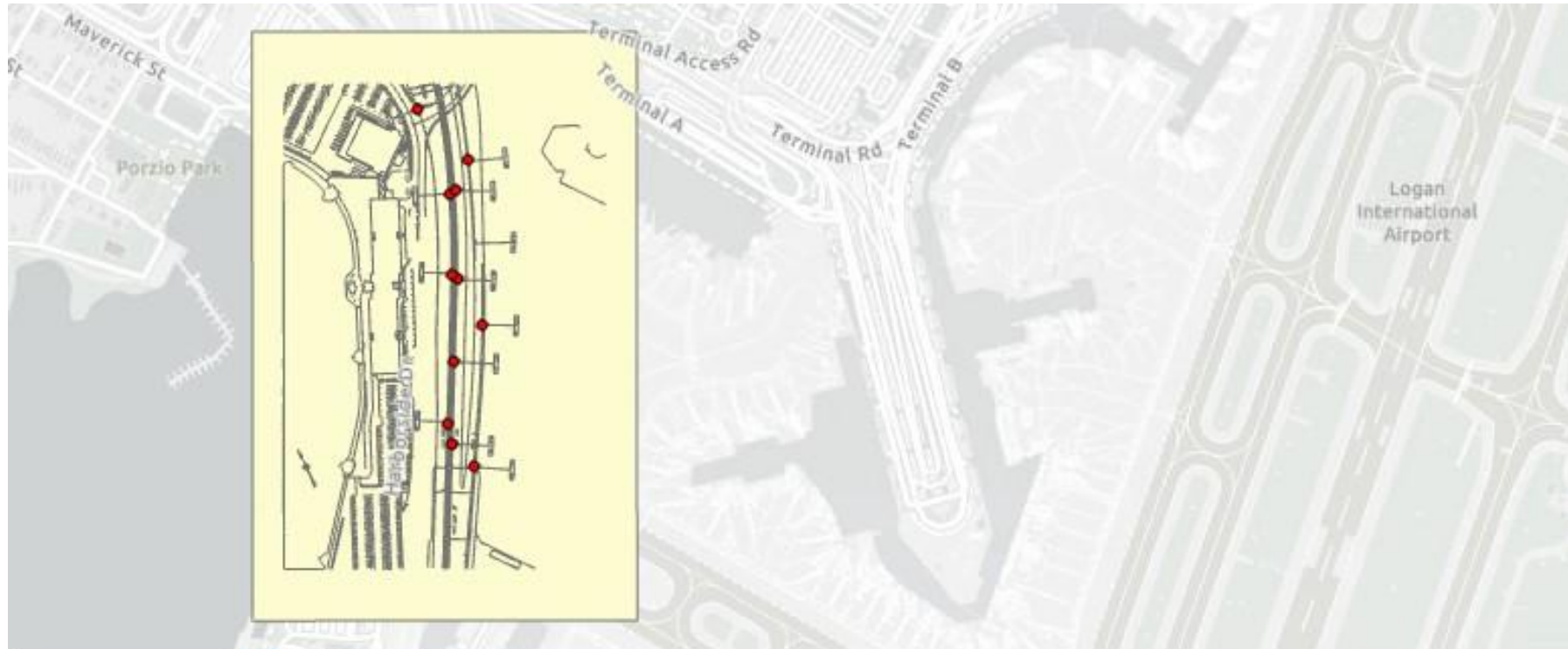
Training Model



Detection Run







1:8,442



71.0157223°W 42.3656098°N

sh1_points

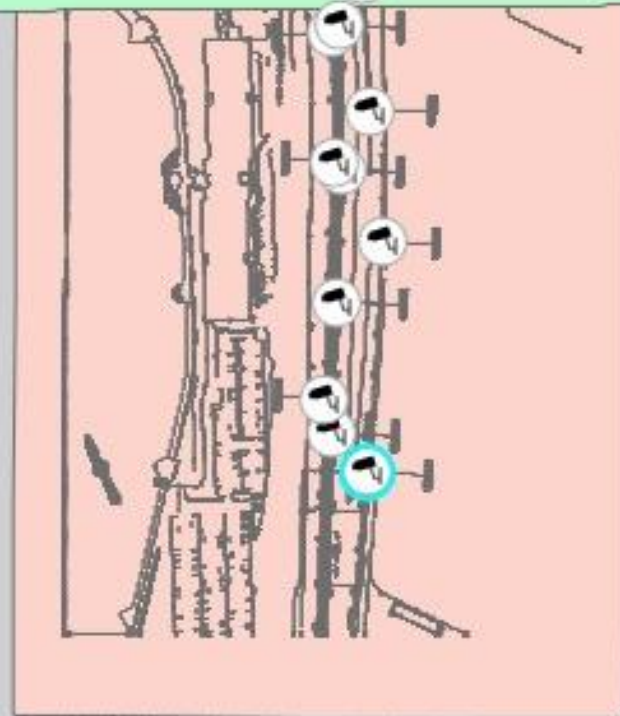
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2	2	Point	1	1	6439	CCTVs	TWT-CCTV-111	42.361939	-71.024734
3	3	Point	1	1	6501	CCTVs	TWT-CCTV-113	42.362145	-71.0248
4	4	Point	1	1	6758	CCTVs	TWT-CCTV-114	42.36276	-71.024694
5	5	Point	1	1	6213	CCTVs	TWT-CCTV-115	42.363126	-71.024319
6	6	Point	1	1	6392	CCTVs	TWT-CCTV-116	42.363581	-71.024655
7	7	Point	1	1	6504	CCTVs	TWT-CCTV-117	42.36363	-71.024732

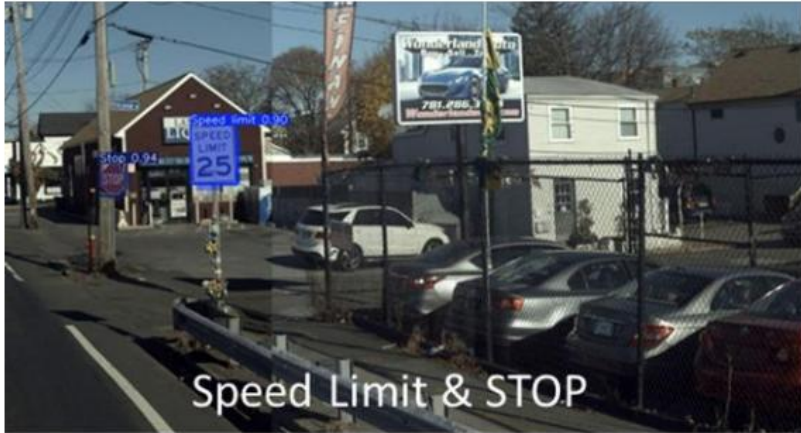
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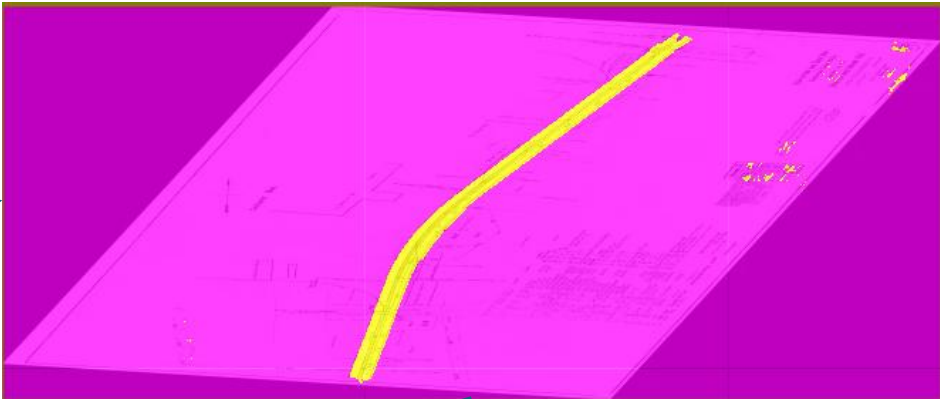
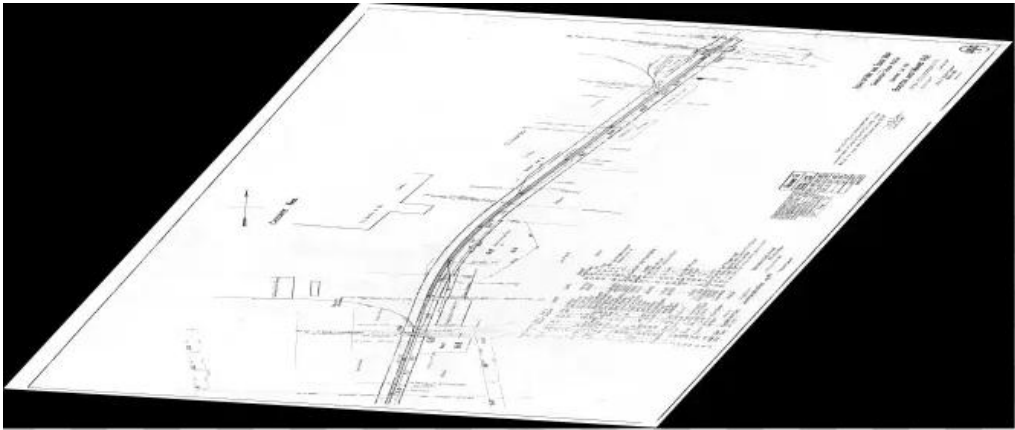


Sample Traffic Sign Detection





Rail right of Way extraction from engineering design plans



Exploring further use cases

Rail

Signal and Equipment Inspection	Monitor signal heads, switch machines, relay houses, communication equipment, and crossing electronics for maintenance needs.
Right-of-Way Encroachment Detection	Detect unauthorized structures, vehicles, pedestrians, fallen trees, landslides, or debris within the rail corridor.
Drainage Monitoring	Identify blocked culverts, ditch erosion, standing water, washouts, and embankment instability that could affect track integrity.

Road

Pavement Marking Inventory	Detect lane markings, painted turn restrictions, stop bars, and pavement symbols while measuring visibility and wear. <u>Useful for routability to determine turn exclusions</u>
Guardrail and Barrier Inventory	Identify guardrails, cable barriers, concrete barriers, attenuators, and end treatments while assessing damage after incidents.



Thanks!



Charles.major@dot.state.ma.us



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Charles.major@dot.state.ma.us



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Charles.major@dot.state.ma.us

